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Leibnitz's Rule for Differentiating Integrals

Thm: Assume that f and $f_{,2}$ (partial derivative) are continuous and that u, v are differentiable. Let

$$G(x) = \int_{u(x)}^{v(x)} f(t, x) dt \quad \text{Then}$$

G is differentiable and

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$$G'(x) = f(v(x), x) v'(x) - f(u(x), x) u'(x) \\ + \int_{u(x)}^{v(x)} f_2(t, x) dt$$

$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(t, x) dt = f(v(x), x) v'(x) - f(u(x), x) u'(x) \\ + \int_{u(x)}^{v(x)} \frac{\partial}{\partial x} f(t, x) dt$$

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Important Special Case

$$\frac{d}{dx} \int_a^b f(t, x) dt = \int_a^b \frac{\partial}{\partial x} f(t, x) dt$$

Rmk: Works for integrals

$$\int_a^\infty f(t, x) dt, \int_{-\infty}^b f(t, x) dt, \int_{-\infty}^\infty f(t, x) dt$$

provided $f(t, x)$, $\frac{\partial}{\partial x} f(t, x)$ decay

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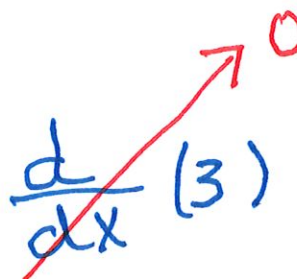
rapidly enough as $t \rightarrow \pm \infty$.

(Be careful with "fat tails".)

Ex: a)
$$\begin{aligned} \frac{d}{dx} \int_1^2 \ln(t^2 + x^2) dt \\ &= \int_1^2 \frac{\partial}{\partial x} \ln(t^2 + x^2) dt \\ &= \int_1^2 \frac{2x}{t^2 + x^2} dt \end{aligned}$$

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$$b) \frac{d}{dx} \int_{\sqrt{x}}^3 e^{-x^2 t^2} dt$$

$$= (\quad) \frac{d}{dx} (3)$$


$$- e^{-x^2 (\sqrt{x})^2} \frac{d}{dx} \sqrt{x} + \int_{\sqrt{x}}^3 \frac{\partial}{\partial x} (e^{-x^2 t^2}) dt$$

$$= -e^{-x^3} \left(\frac{1}{2\sqrt{x}} \right) + \int_{\sqrt{x}}^3 (-2xt^2 e^{-x^2 t^2}) dt$$

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1st Order ODE

Ordinary Differential Equations

I. Separable, II. Linear

III. Substitution to make an
eqn separable or linear.

⑦

Separation of Variables

$$y'(t) = f(t)g(y(t))$$

Assume $g(y) \neq 0$

$$\frac{y'(t)}{g(y(t))} = f(t)$$

Choose H s.t. $H'(y) = \frac{1}{g(y)}$

F s.t. $F'(t) = f(t)$

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$$\begin{aligned}\frac{d}{dt} H(y(t)) &= H'(y(t)) y'(t) \\ &= \frac{y'(t)}{g(y(t))}\end{aligned}$$

$$\frac{d}{dt} H(y(t)) = \frac{d}{dt} F(t)$$

↙ constant

$$H(y(t)) = F(t) + C$$

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Standard to write

$$\frac{dy}{dt} = f(t)g(y)$$

$$\frac{dy}{g(y)} = f(t)dt$$

$$\int \frac{dy}{g(y)} = \int f(t)dt + \text{Constant}$$

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Ex: $y'(t) = t^3 e^{y(t)}$

$$\frac{y'(t)}{e^{y(t)}} = t^3$$

$$H'(y) = \frac{1}{e^y} = e^{-y}$$

$$H(y) = -e^{-y}$$

$$F(t) = \frac{1}{4} t^4$$

$$-e^{-y(t)} = \frac{1}{4} t^4 + C$$

⑪

1st Order Linear

$p, q: (a, b) \rightarrow \mathbb{R}$ continuous

$t_0 \in (a, b)$ given, $y_0 \in \mathbb{R}$ given

Diff Egn $y'(t) + p(t)y(t) = q(t)$

$y(t_0) = y_0$ Initial Condition

Like $\frac{d}{dt} (\quad) = \text{Fn of } t$

Product Rule

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$$\frac{d}{dt}(\mu y(t)) = \mu y'(t) + \square y(t)$$

Look for $\mu(t)$ (Integrating Factor)

s.t. $\mu(t) \neq 0$ and

$$\mu(t)y'(t) + \mu(t)p(t)y(t)$$

$$\stackrel{\text{want}}{=} \frac{d}{dt}(\mu(t)y(t))$$

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Need

$$\begin{aligned}\mu(t)y'(t) + \mu(t)p(t)y(t) \\ = \mu(t)y'(t) + \mu'(t)y(t)\end{aligned}$$

$$\mu'(t) = \mu(t)p(t)$$

$$\frac{\mu'(t)}{\mu(t)} = p(t)$$

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W.l.o.g. $\mu(t) > 0$

$$\ln(\mu(t)) = \int p(t) dt$$

Choose P s.t. $P'(t) = p(t)$

$$\mu(t) = \exp[P(t)]$$

$$\mu(t) y'(t) + \underbrace{\mu(t) p(t)}_{\mu'(t)} y(t) = \mu(t) q(t)$$

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$$\frac{d}{dt}(\mu(t)y(t)) = \mu(t)q(t)$$

$$\frac{d}{ds}(\mu(s)y(s)) = \mu(s)q(s) \quad \left[\int_{t_0}^t \right]$$

$$\mu(s)y(s) \Big|_{t_0}^t = \int_{t_0}^t \mu(s)q(s)ds$$

$$\mu(t)y(t) - \mu(t_0)y(t_0) = \int_{t_0}^t \mu(s)q(s)ds$$

$$\text{Ex: } y'(t) + 6y(t) = e^{-2t}, \quad y(0) = 4 \quad (16)$$

$$p(t) = 6, \quad P(t) = 6t, \quad \mu(t) = e^{6t}$$

$$\underbrace{e^{6t} y'(t) + 6e^{6t} y(t)} = e^{4t}$$

$$\frac{d}{dt}(e^{6t} y(t)) = e^{4t}$$

$$e^{6t} y(t) = \frac{1}{4} e^{4t} + C$$

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Find C. Put $t=0$

$$e^0(4) = \frac{1}{4}e^0 + C$$

$$C = \frac{15}{4}$$

$$e^{6t}y(t) = \frac{1}{4}e^{4t} + \frac{15}{4}$$

$$y(t) = \frac{1}{4}e^{-2t} + \frac{15}{4}e^{-6t}$$

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$$\text{Ex: } t y'(t) = 3y(t) + t^3$$

$$y(1) = -1$$

$$\textcircled{*} \quad y'(t) - \frac{3}{t} y(t) = t^2 \quad t > 0$$

$$p(t) = -\frac{3}{t} \quad P(t) = -3 \ln t$$

$$\mu(t) = \exp[-3 \ln t] = t^{-3}$$

$$t^{-3} y'(t) - 3t^{-4} y(t) = t^{-1}$$

$$\frac{d}{dt} (t^{-3} y(t)) = \frac{1}{t}, \quad t > 0$$

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$$t^{-3}y(t) = \ln t + C$$

$$\text{Put } t=1$$

$$-1 = \ln 1 + C \quad C = -1$$

$$t^{-3}y(t) = \ln t - 1$$

$$y(t) = t^3 \ln t - t^3$$

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2nd Order Linear ODEs

$$(*) \quad A(t)y''(t) + B(t)y'(t) + C(t)y(t) = g(t)$$

A, B, C, g cont. on I - interval

Need $A(t) \neq 0$ all $t \in I$

$$(H) \quad A(t)y''(t) + B(t)y'(t) + C(t)y(t) = 0$$

↑ homogeneous

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Superposition

1. If y_1, y_2 solns of (H) and c_1, c_2 constants then $c_1 y_1 + c_2 y_2$ is a soln of (H)
2. If y_1, y_2 solns of (H) and y_p is a soln of $\textcircled{*}$ then $c_1 y_1 + c_2 y_2 + y_p$ is a soln of $\textcircled{*}$
3. If y_{p1}, y_{p2} solns of $\textcircled{*}$ then $y_{p1} - y_{p2}$ is a soln of (H)

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Two Special Cases where
we can generate all solns of (H)

I. A, B, C Constant

$$A(t) = a, \quad B(t) = b, \quad C(t) = c$$

$$(1) \quad ay''(t) + by'(t) + cy(t) = 0$$

Look at Quadratic Eqn

$$(C1) \quad a\lambda^2 + b\lambda + c = 0 \quad (\text{Plug } y = e^{\lambda t} \text{ into (1)})$$

(i) Distinct Real Roots λ_1, λ_2

$$y = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$$

(ii) Double Real Root λ

$$y = c_1 e^{\lambda t} + c_2 t e^{\lambda t}$$

(iii) Complex Conjugate Roots $\lambda = \alpha + \beta i$
 α, β real ($\beta \neq 0$), $i^2 = -1$

$$y = e^{\alpha t} (c_1 \cos \beta t + c_2 \sin \beta t)$$

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$$\text{II. } A(t) = at^2, B(t) = bt, C(t) = c$$

$$(2) \quad at^2 y''(t) + bt y'(t) + cy(t) = 0, \quad t > 0$$

$$\text{Let } Z = \ln t$$

$$\frac{dy}{dt} = \frac{dy}{dz} \left(\frac{1}{t} \right), \quad \frac{d^2 y}{dt^2} = \frac{dy}{dz} \left(-\frac{1}{t^2} \right) + \frac{d^2 y}{dz^2} \left(\frac{1}{t^2} \right)$$

$$at^2 \left[\frac{1}{t^2} \frac{d^2 y}{dz^2} - \frac{1}{t^2} \frac{dy}{dz} \right] + bt \frac{dy}{dz} \left(\frac{1}{t} \right) + cy = 0$$

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$$a \frac{d^2 y}{dz^2} + (b-a) \frac{dy}{dz} + cy = 0$$

$$(C2) \quad a\lambda^2 + (b-a)\lambda + c = 0$$

(i) Distinct real roots λ_1, λ_2

$$\begin{aligned} y &= c_1 e^{\lambda_1 z} + c_2 e^{\lambda_2 z} \\ &= c_1 e^{\lambda_1 \ln t} + c_2 e^{\lambda_2 \ln t} \\ &= c_1 t^{\lambda_1} + c_2 t^{\lambda_2} \end{aligned}$$

(ii) Double Real Root λ

$$y = c_1 e^{\lambda z} + c_2 z e^{\lambda z} \\ = c_1 t^{\lambda} + c_2 (\ln t) t^{\lambda}$$

(iii) Complex Conjugates
 $\lambda = \alpha \pm \beta i \quad \beta \neq 0$

$$y = e^{\alpha z} (c_1 \cos(\beta z) + c_2 \sin(\beta z)) \\ = t^{\alpha} (c_1 \cos(\beta \ln t) + c_2 \sin(\beta \ln t))$$

Ex 1: $y'' + 6y' + 9y = 0$

$$\lambda^2 + 6\lambda + 9 = 0 \quad (\lambda + 3)^2 = 0$$

$$\lambda = -3 \quad \text{Double}$$

$$y = c_1 e^{-3t} + c_2 t e^{-3t}$$

Ex 2: $y'' + 16y = 0 \quad \lambda^2 + 16 = 0$

$$\lambda^2 = -16 \quad \lambda = \pm 4i$$

Conjugates $\alpha = 0, \beta = 4$

$$y = c_1 \cos 4t + c_2 \sin 4t$$

Ex 3: $y'' - 25y = 0$

$$\lambda^2 - 25 = 0 \quad (\lambda - 5)(\lambda + 5) = 0$$

$$\lambda_1 = 5, \quad \lambda_2 = -5$$

$$y = c_1 e^{5t} + c_2 e^{-5t}$$

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Black-Scholes ODE

For an option that doesn't have an expiration date

Price is $v(x)$ when stock price is x

$$\frac{1}{2} \sigma^2 x^2 v''(x) + r x v'(x) - r v(x) = 0$$

Form (2) with $a = \frac{1}{2} \sigma^2$, $b = r$,

$$c = -r$$

$$\frac{1}{2}\sigma^2\lambda^2 + (r - \frac{1}{2}\sigma^2)\lambda - r = 0$$

$$\lambda = \frac{-(r - \frac{1}{2}\sigma^2) \pm \sqrt{(r - \frac{1}{2}\sigma^2)^2 + 2r\sigma^2}}{\sigma^2}$$

$$= \frac{-(r - \frac{1}{2}\sigma^2) \pm \sqrt{(r + \frac{1}{2}\sigma^2)^2}}{\sigma^2}$$

$$\lambda_1 = 1, \quad \lambda_2 = \frac{-2r}{\sigma^2}$$

$$V(x) = C_1 x + C_2 x^{-2r/\sigma^2}$$

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Black-Scholes PDE

$v(t, x)$ = price at time t of
a derivative security
given that the stock
price at t is x .

$$\frac{\partial v}{\partial t} + \frac{1}{2} \sigma^2 x^2 \frac{\partial^2 v}{\partial x^2} + r x \frac{\partial v}{\partial x} - r v = 0$$

You are not responsible for
this in Math Prep.